Exercises: Statistical Testing and Confidence Regions

Mathematical Foundations for Machine Learning

Statistical Testing Fundamentals

Exercise 1.1 [Easy]

A medical test is used to detect a disease. Let H_0: "patient is healthy" and H_1: "patient is sick." The test is applied to 200 patients and produces the following results:

	Test says Healthy	Test says Sick
Truly Healthy	130	20
Truly Sick	10	40

(a) Identify the Type I error and the Type II error in this context. Which one corresponds to a false positive?

(b) Compute the significance level (Type I error rate) and the Type II error rate from the table.

(c) Compute the power of this test. Interpret the result in one sentence. $\beta(\theta) = P_\theta(\text{reject } H_0)$

power = 1 - type2 = 1 - P(test says healthy | truly sick) = P(test says sick | truly sick)

Exercise 1.2 [Easy]

Let $X_1, \ldots, X_n$ be i.i.d. samples from a distribution with mean μ and known variance σ^2. We construct the confidence interval

$$P_\mu\left(\left|\sqrt{n}\frac{\hat{\mu}-\mu}{\sigma}\right| \le q_{1-\alpha/2}\right) = 1-\alpha$$

$$C_n = \left[\bar{X}_n - \frac{\sigma}{\sqrt{n}}q_{1-\alpha/2},\ \bar{X}_n + \frac{\sigma}{\sqrt{n}}q_{1-\alpha/2}\right],$$

where $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ and q_γ is the γ-quantile of $\mathcal{N}(0,1)$.

Definition of quantile (|Z|) -> P(Z<q/2) + P(

$-q_{1-\alpha/2} \le \frac{\hat{\mu}-\mu}{\sigma/\sqrt{n}} \le q_{1-\alpha/2}$

$\hat{\mu} - \frac{\sigma}{\sqrt{n}}q_{1-\alpha/2} \le \mu \le \hat{\mu} + \frac{\sigma}{\sqrt{n}}q_{1-\alpha/2}$

(a) State precisely: what is random in C_n, and what is fixed? Is the statement "μ falls in C_n with probability $1-\alpha$" correct?

(b) Suppose $\sigma = 2$, $\alpha = 0.05$, and $q_{0.975} = 1.96$. How large must n be so that the width of the confidence interval is at most 0.5?

(c) A colleague claims: "If I compute a 95% confidence interval and the parameter μ is not in it, then my experiment failed." Is this interpretation correct? Explain briefly.

Exercise 1.3 [Medium] (That is gaussian test)

A classifier is evaluated on a test set of n samples. Each sample is independently classified correctly with unknown probability p (the true accuracy). Let $\hat{p} = \frac{1}{n}\sum_{i=1}^n Z_i$ be the empirical accuracy, where $Z_i \in \{0,1\}$ indicates whether sample i was correctly classified.

(a) Compute $\mathbb{E}[\hat{p}]$ and $\mathrm{Var}(\hat{p})$.

(b) Using Chebyshev's inequality, show that for any $\varepsilon > 0$,

$$P(|\hat{p} - p| \ge \varepsilon) \le \frac{1}{4n\varepsilon^2}.$$

Hint: What is the maximum of $p(1-p)$ over $p \in [0,1]$?

(c) Using the result from (b), how many test samples n are needed to guarantee that $|\hat{p}-p| < 0.05$ with probability at least 95%?

Exercise 1.4 [Medium]

Consider a hypothesis test with null hypothesis $H_0 : \theta = \theta_0$ and a test statistic $T = T(X_1, \ldots, X_n)$ whose distribution under H_0 is known. The p-value is defined as

$$T = \frac{\text{estimator} - \text{value claimed by } H_0}{\text{standard deviation of the estimator}}$$

$$\text{p-value} = P_{H_0}(T \geq t_{\text{obs}}),$$

test statistic -> how many std. dev. away from H0

where t_{obs} is the observed value of T.

(a) A researcher runs a test and obtains a p-value of 0.03. The significance level is $\alpha = 0.05$. State the decision rule and whether H_0 is rejected.

(b) The researcher repeats the same experiment independently 5 times and obtains p-values: $0.03, 0.12, 0.07, 0.04, 0.15$. If the significance level is $\alpha = 0.05$ for each individual test, a colleague argues that at least one rejection is "proof" that H_0 is false. Show that even if H_0 is true, the probability of at least one rejection among m independent tests at level α equals $1 - (1 - \alpha)^m$, and compute this for $m = 5$, $\alpha = 0.05$.

(c) Explain how the Bonferroni correction addresses the issue in (b). What adjusted significance level should be used for each of the 5 tests to control the overall false rejection probability at $\alpha = 0.05$?

Exercise 1.5 [Hard]

Let T be a continuous test statistic and define the p-value as $p = P_{H_0}(T \geq T_{\text{obs}})$, where T_{obs} is the observed realization of T.

(a) Let F_0 denote the CDF of T under H_0. Show that the p-value can be written as $p = 1 - F_0(T_{\text{obs}})$.

(b) Prove that under H_0, the p-value is uniformly distributed on $[0, 1]$, i.e., $p \sim \text{Uniform}(0, 1)$.

Hint: Use the probability integral transform: if T has CDF F_0 and F_0 is continuous, then $F_0(T) \sim \text{Uniform}(0, 1)$.

(c) Using the result of (b), show that a test that rejects H_0 when $p \leq \alpha$ has exactly significance level α, i.e., $P_{H_0}(p \leq \alpha) = \alpha$.

Confidence Intervals and the Gauss Test

Exercise 2.1 [Easy]

Consider i.i.d. samples $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ with known σ^2. We wish to test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$ using the Gauss test statistic

X follows normal, Z follows N(0, 1)

$$Z = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma}.$$

(a) State the distribution of Z under H_0.

(b) For a two-sided test at significance level α, state the rejection region in terms of Z and the quantile $q_{1-\alpha/2}$ of $\mathcal{N}(0, 1)$.

(c) Suppose $n = 25$, $\sigma = 3$, $\bar{X}_n = 5.2$, $\mu_0 = 4.5$, and $q_{0.975} = 1.96$. Compute Z and state whether H_0 is rejected at $\alpha = 0.05$.

Exercise 2.2 [Medium]

Consider the Gauss test setting: $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d. with known σ^2, and the two-sided test $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$ at level α.

(a) Show that the test "reject H_0 if $|Z| > q_{1-\alpha/2}$" is equivalent to "reject H_0 if $\mu_0 \notin C_n$", where

$$C_n = \left[\bar{X}_n - \frac{\sigma}{\sqrt{n}} q_{1-\alpha/2},\ \bar{X}_n + \frac{\sigma}{\sqrt{n}} q_{1-\alpha/2}\right]$$

is the $(1 - \alpha)$-confidence interval for μ.

(b) This shows that hypothesis tests and confidence intervals are two sides of the same coin. Formulate the general principle: a value θ_0 is rejected by a level-α test if and only if $\theta_0 \notin \ldots$ (complete the statement).

(c) Suppose we are comparing two classifiers A and B on a test set of n samples. Classifier A has accuracy $\hat{p}_A = 0.82$ and classifier B has $\hat{p}_B = 0.79$, each evaluated on independent test sets of size $n = 400$. Using the normal approximation, construct a 95% confidence interval for the difference $p_A - p_B$. Does the interval contain 0? Interpret the result.

Exercise 2.3 [Hard]

$\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$

The power function of a test is defined as $\beta(\theta) = P_\theta(\text{reject } H_0)$.

Consider the one-sided Gauss test: $X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2)$ i.i.d. with known σ^2, testing $H_0 : \mu \leq \mu_0$ vs. $H_1 : \mu > \mu_0$. The test rejects H_0 if $Z = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma} > q_{1-\alpha}$.

(a) Derive the power function $\beta(\mu)$ for this test. Express it in terms of the standard normal CDF Φ.

Hint: Under μ, write $Z = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} + \frac{\sqrt{n}(\mu - \mu_0)}{\sigma}$.

(b) Show that $\beta(\mu_0) = \alpha$ and that $\beta(\mu) \to 1$ as $\mu \to \infty$. What does this mean about the test?

(c) Suppose $\sigma = 1$, $\alpha = 0.05$ (so $q_{0.95} \approx 1.645$), and we want the test to have power at least 0.9 when $\mu = \mu_0 + 0.5$. Find the minimum sample size n.

Hint: $\Phi^{-1}(0.9) \approx 1.282$.

Permutation Tests

Exercise 3.1 [Easy]

In permutation test you don't divide by std. bec. both distributions are on the same data (or should)

Suppose we have two groups of observations: $X_1 = 3, X_2 = 7$ and $Y_1 = 10$. We test $H_0 : F_X = F_Y$ (the two groups come from the same distribution) using the test statistic $T = |\bar{X} - \bar{Y}|$.

(a) Compute the observed test statistic t_{obs}.

(b) List all $\binom{3}{2} = 3$ ways to split the pooled data $\{3, 7, 10\}$ into a group of size 2 and a group of size 1, and compute T for each split.

permutations w.r.t. which group each element

(c) Compute the p-value, i.e., $P_0(T \geq t_{\text{obs}})$, by counting the fraction of permutations where $T \geq t_{\text{obs}}$. Can we reject H_0 at $\alpha = 0.05$?

Exercise 3.2 [Medium]

In practice, the exact permutation test is intractable for large datasets because the number of permutations $\binom{N}{m}$ grows combinatorially. Instead, one uses a **Monte Carlo approximation**: sample S random permutations and estimate

$$\hat{p} = \frac{1}{S} \sum_{j=1}^{S} \mathbf{1}_{T_j \geq t_{\text{obs}}}.$$

(a) Show that $\hat{p}$ is an unbiased estimator of the true p-value $p^* = P_0(T \geq t_{\text{obs}})$.

(b) Compute $\mathrm{Var}(\hat{p})$ and show it is at most $\frac{1}{4S}$.

(c) Using Chebyshev's inequality, how many Monte Carlo samples S are needed to guarantee $|\hat{p} - p^*| < 0.01$ with probability at least 0.99?

Mixed Challenging Exercises

Exercise 4.1 [Hard]

A machine learning practitioner trains a binary classifier and evaluates it on a test set of $n = 500$ i.i.d. samples. The classifier achieves an empirical accuracy of $\hat{p} = 0.74$.

(a) The practitioner wants to test $H_0 : p \leq 0.70$ (the classifier is no better than 70% accuracy) against $H_1 : p > 0.70$. Using the normal approximation under H_0 with worst-case variance (i.e., $p_0 = 0.70$), compute the test statistic Z and decide at $\alpha = 0.05$ ($q_{0.95} \approx 1.645$).

(b) Construct a 95% confidence interval for p using the normal approximation with estimated variance $\hat{p}(1-\hat{p})/n$. Verify that the decision from part (a) is consistent with whether 0.70 lies inside or outside this interval.

(c) The practitioner now evaluates the same classifier on $k = 10$ different test sets (e.g., different demographic subgroups) and tests $H_0^i : p_i \leq 0.70$ for each. Explain why a Bonferroni correction is appropriate and compute the adjusted significance level. If the smallest p-value across the 10 tests is 0.008, is any subgroup's performance significantly above 70%?

Exercise 4.2 [Hard]

Let $X_1, \ldots, X_n$ be i.i.d. with $X_i \in [0, 1]$ (e.g., bounded loss values of a model). Let $\mu = \mathbb{E}[X_i]$ denote the true expected loss.

(a) Using Hoeffding's inequality, show that

$$P\big(|\bar{X}_n - \mu| \geq t\big) \leq 2\exp(-2nt^2).$$

Recall Hoeffding: For independent $X_i \in [a_i, b_i]$, $P\big(|\bar{X}_n - \mu| \geq t\big) \leq 2\exp\Big(\frac{-2n^2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\Big)$.

(b) Invert the inequality from (a) to construct a $(1-\alpha)$-confidence interval for μ that does not require knowledge of the variance. Express the interval width as a function of n and α.

(c) Compare the width of this Hoeffding-based confidence interval to the normal-approximation confidence interval $\bar{X}_n \pm \frac{\sigma}{\sqrt{n}} q_{1-\alpha/2}$ (assuming σ known). For which regime (small n or large n) is each preferable, and why?